Prashant Dilip Desai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.96
Kirloskar Junior College, Kirloskarwadi	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education, Pune)	2015	74.46%
Swami Ramanand Vidyalaya, Ramandnagar	Class X (Maharashtra State Board of Secondary and Higher Secondary Education, Pune)	2013	89.82%

EXPERIENCE

IIT, Hyderabad

Research Assistant

- Worked on self-interference cancellation using a Vector Modulator in a full-duplex radio.

KPIT Technologies, Bangalore

System Test Engineer

- Developed and executed CANoe-based simulations and test automation using CAPL scripting for system-level validation.
- Performed system testing of ADAS front-camera (ADCAM) and LiDAR modules, including validation of UDS, automotive Ethernet, and cybersecurity features.

PROJECTS

SPI control of MCP4912 (10-bit 2-channel DAC) from scratch

- Implemented serial data transmission and timing control using Verilog HDL on the ZCU102 platform.
- Generated programmable 10-bit DAC output codes for analog output control based on user-defined input values.
- Verified SPI communication and DAC control through RTL simulation and hardware implementation on the ZCU102 platform.

Low-Pass Filter Design of 1.92MHz using Filter Coefficients extracted from MATLAB

- Converted floating-point filter coefficients into hardware-compatible integer values for FPGA-based FIR filter implementation.
- Implemented the FIR filter architecture using Verilog HDL for low-pass digital signal processing applications.
- Verified filter operation using in-band and out-of-band signals through RTL simulation and frequency response analysis.

Binary Coded Delay Element with variable data width for Matching delays of two RF paths

- Implemented a controllable delay element with input-based delay enable and disable functionality for RF path matching.
- Cascaded delay elements with individually configurable delays based on successive powers of two.
- Verified programmable delay operation through RTL simulation and hardware implementation using the ZCU102

FPGA platform.

PCB design of an attenuator-based Vector Modulator and a 4-quadrant Vector Modulator

- Designed 4-layer RF PCB layouts for an attenuator-based and a 4-quadrant vector modulator.
- Implemented I/Q signal paths for independent amplitude and phase control of the input RF signal.
- Designed RF, control, and power interfaces for vector-modulator operation and system-level hardware integration.
- Optimized component placement and high-frequency signal routing for RF performance and designed matching circuitry.

Two-Stage Miller-Compensated Op-Amp (TSMC 180 nm)

- Designed and simulated a two-stage CMOS op-amp with Miller compensation for high gain and stability

Automatic Car Washing System Using PLC

- Designed an automatic car washing system comprising 3 compartments: Drying, Washing and Foaming, Brushing
- Implemented a threaded bolt mechanism for moving the car through compartments using a DC motor.
- Integrated IR sensors to detect the position of the car in each compartment and initiate the automatic cleaning process.
- Designed relay-based interfaces for signal conversion and isolation between sensors, PLCs and actuators.

PUBLICATIONS

Real-time Tuning for RF domain Self Interference Cancellation in Full-Duplex Radio

- Developed an adaptive RF-domain Self-Interference Cancellation system using EBD and a Vector Modulator.
- Combined an Electrical Balanced Duplexer and a Vector Modulator for precise amplitude and phase alignment of Tx leakage.
- Implemented Coordinate-Wise Adaptive Search for iterative EBD and Vector Modulator tuning.
- Demonstrated up to 50 dB of Self-Interference Cancellation at 15 dBm Tx power with antenna interaction.

RF domain Self Interference Cancellation using Vector Modulator in UHF band

- Developed a wideband FPGA-controlled vector modulator for RF-domain self-interference cancellation.
- Integrated a quadrature hybrid, dual-channel vector multiplier, and balun for self-interference suppression.
- Implemented programmable I/Q control using dual-channel 10-bit DACs driven by ZCU102.
- Demonstrated over 36 dB analog Self Interference Cancellation across 225-400 MHz using an ADRV9002 transceiver.

SCHOLASTIC ACHIEVEMENTS

Best Departmental Project Award in B.E.

SKILLS

Tools: Xilinx Vivado, Cadence Virtuoso, ADS, KiCad, MATLAB

RELEVANT COURSES

Analog Domain: Analog IC Design, Integrated Circuits for Wireless Communication, Analog IC Design Lab

Digital Domain: Digital IC Design, Memory Circuits and System Design

Technology & Devices: VLSI Technology, Semiconductor Device Modelling, Electronic Package Design

POSITIONS OF RESPONSIBILITY

Teaching Assistant (TA) for NPTEL course: IC design for Wireless Communication

Managed key course logistics. **Instructor:** Dr. Abhishek Kumar

Teaching Assistant (TA): IC Design for Wireless Communication Course for 4th-year UG and 1st-year PG students

Managed key course logistics. **Instructor:** Dr. Abhishek Kumar

Teaching Assistant (TA): Analog IC Design Course for 3rd-year UG and 1st-year PG students

Managed key course logistics. **Instructor:** Dr. Abhishek Kumar

EXTRACURRICULAR

Cooking traditional regional recipes - Anime enthusiast - Reading - Calisthenics and bodyweight training